



HighTech Innovators

YOUR SOFTWARE. BETTER

DevOps++

*Embedding Sustainability into
Everyday Software Engineering*



Green Software
CHAMPIONS



Green
Software
Foundation

What is Green Coding?

Understanding the unexpected approach to green coding.

Moving beyond code libraries and instant line optimizations.

```
How much J-s  
would this function be  
public bool AreWeThereYet()  
{ return false; }?
```

CNIT FOREST

Green IQ
powered by e

The place for responsible
greening your day

z

z

z

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z

z

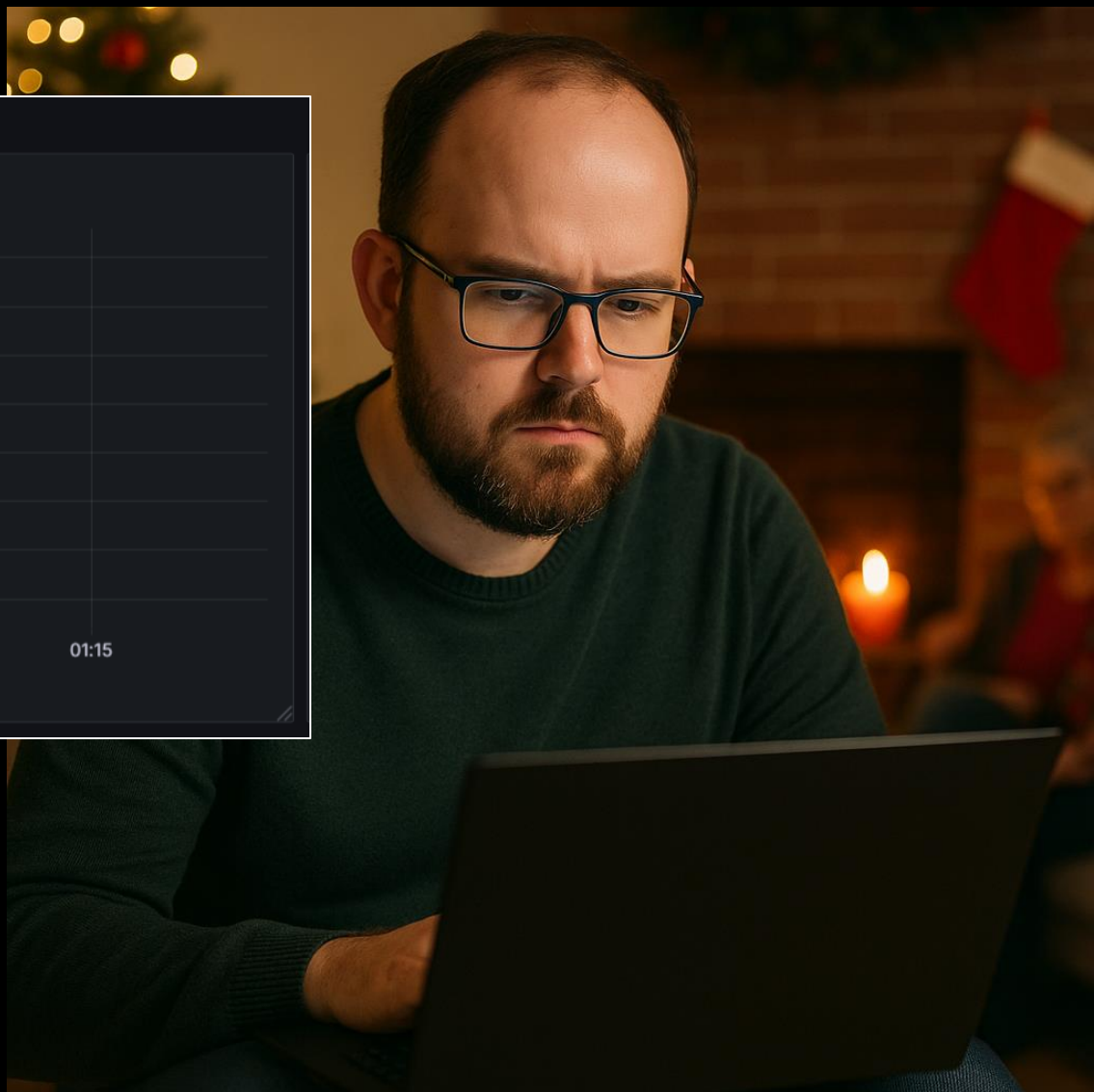
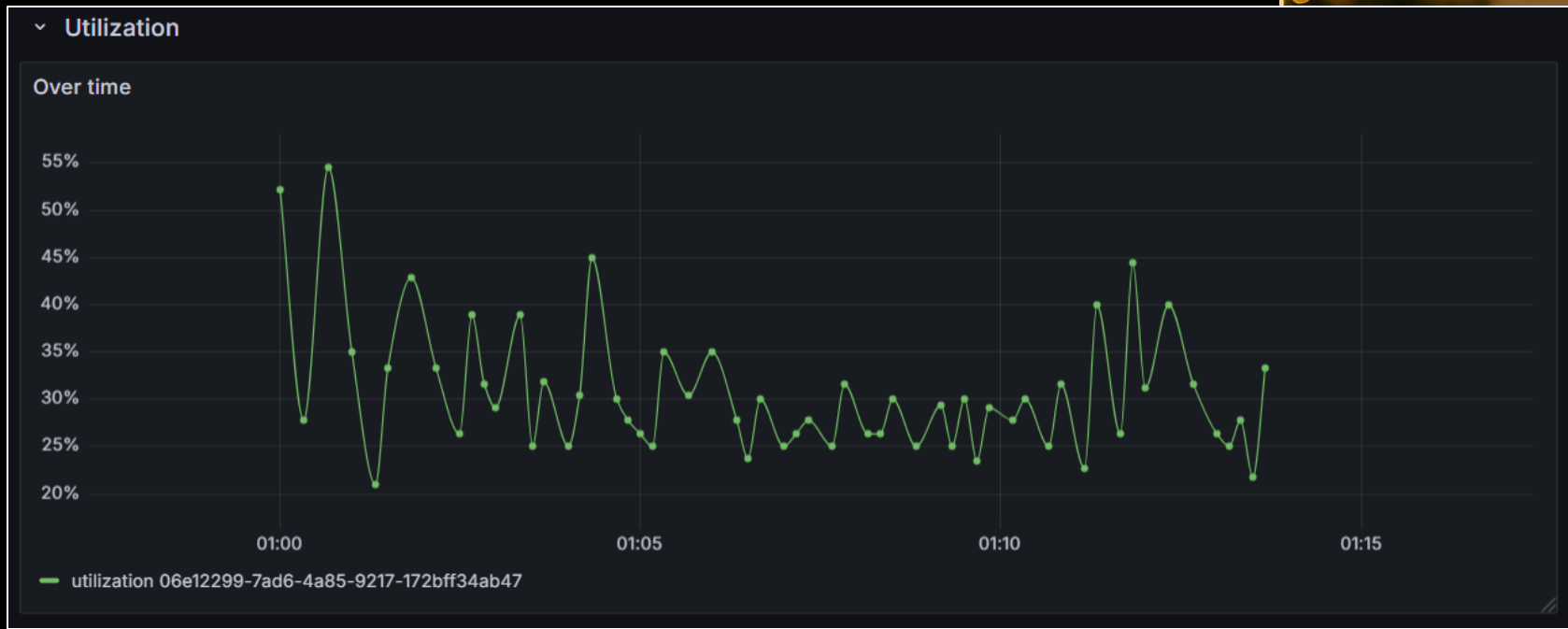
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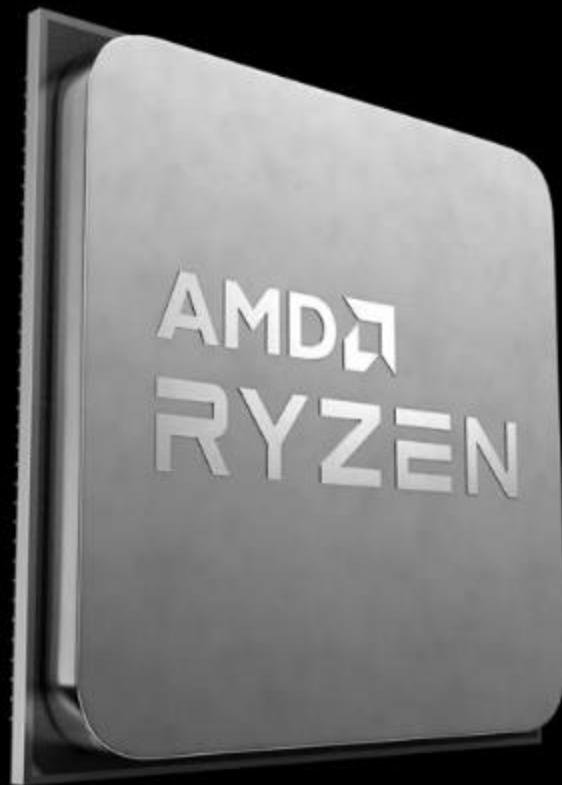
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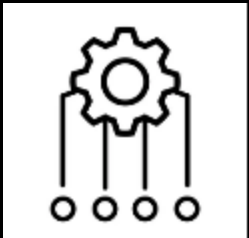
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Cracking the Code of Green and Sustainable IT

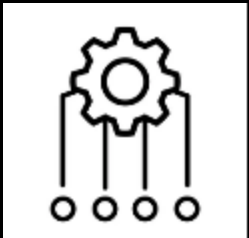








5 Ghz	5 Ghz	Idle
Idle	Idle	Idle



**5
Ghz**

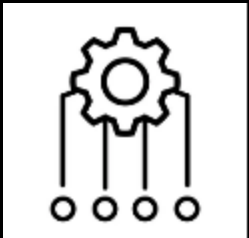
**5
Ghz**

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Ghz**

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Ghz**

**5
Ghz**



**3.6
Ghz**

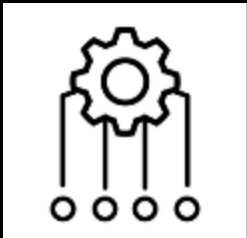
**3.6
Ghz**

**3.6
Ghz**

**3.6
Ghz**

**3.6
Ghz**

**3.6
Ghz**



**CPU Package
(example
25 watts)**

**1.0
volts**

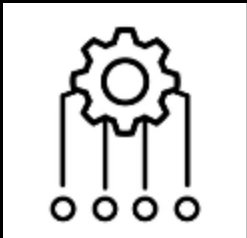
**1.0
volts**

**1.0
volts**

**1.0
volts**

**1.0
volts**

**1.0
volts**



**CPU Package
(example
25 watts)**

**≈ 4.2
W**

**≈ 4.2
W**

**≈ 4.2
W**

**≈ 4.2
W**

**≈ 4.2
W**

**≈ 4.2
W**

What does
Sustainable IT mean
beyond hardware
efficiency?

Every kWh $\approx$ CO₂

*Even if hardware is “efficient,”
software **inefficiency**
still burns energy*

Search areas



 **Netherlands**

Sep 29, 2025, 12:00 PM GMT+2

277g
CO₂eq/kWh

Carbon Intensity

54%

Low-carbon

51%

Renewable

Preliminary

Sep 29, 2025, 12:00 PM GMT+2

72 Hours

<>

2:00 PM

Sep 27

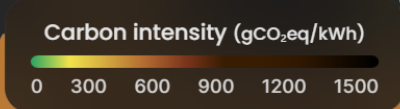
2:00 PM

Sep 28

2:00 PM

Sep 29

Live



 A faint, light green fingerprint graphic is centered in the background of the header section.

MANIFESTO



Minimise Carbon

The Foundation's mission is to reduce the total change in global carbon emissions associated with software. When evaluating choices **we choose the option that advocates for abatement (reducing emissions) not neutralisation (offsetting emissions).**

Operationalise:

We will consider how to **minimise** carbon emissions in every decision we make around how we conduct ourselves operationally and the standards and technology we create and use.

OUR STEERING MEMBERS





41,600,000,000,000,000 grams CO₂e

global CO₂ emissions per year (IPCC/Global Carbon Project)



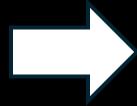
41,600,000,000,000,000 grams CO₂e

global CO₂ emissions per year (IPCC/Global Carbon Project)

12,000,000,000,000 grams CO₂e

Scope 3 ASML emissions in 2024

44.027 employees



32,800,000,000 grams CO₂e

Scope 1 & 2 ASML operational emissions in 2024

81,030,000 grams CO₂e

1000 virtual example servers (100g co₂e / kWh)

220 g CO₂e/kWh

2023 CBS

To answer **what**
Sustainable IT
really solves

Kill daily waste
*Efficient concurrency,
async, IO-aware design.*

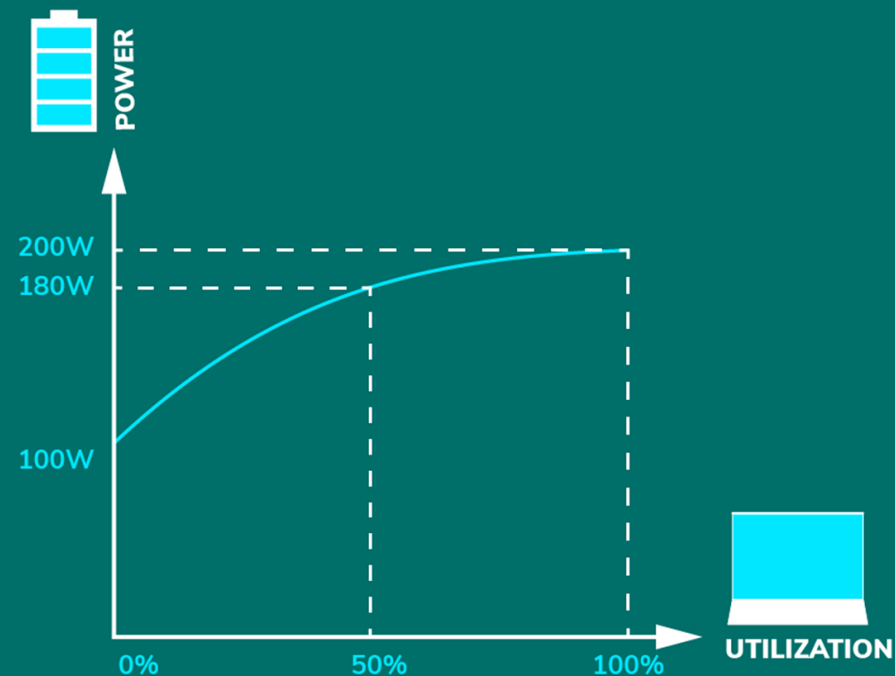
Extend system life

*Modular, Clean, SOLID,
governance-driven code
that adapts instead of
being rewritten.*

Trace CO₂ back to code choices
*Underutilization and
bad design = hidden emissions.*

Trace CO₂ back to code choices

*Underutilization and
bad design = hidden emissions.*



Shift focus from watts to *value per watt*
*Every user action should be served with
minimal operations & energy.*

So, what is the
DevOps++ framework?

What is DevOps?

What is DevOps?

Break Silos

Dev + Ops work as one loop.

Plan → code → build → test → deploy → run → monitor → improve.

What is DevOps?

Automate Flow

CI/CD pipelines, infra as code, observability → faster & safer delivery.

What is DevOps?

Feedback Loop

User metrics + ops signals flow back to dev decisions.

What is DevOps++?

What is DevOps++?

Add Sustainability Lens

Detection of **actions** that will lead to

reduction of **energy** use, **idle** compute,
CO₂ footprint in production.

Extending device **lifetime**
with improved software-based **heat** regulation.

What is DevOps++?

Add Sustainability Lens

Detection of **actions** that will lead to

reduction of **energy** use, **idle** compute,
CO₂ footprint in production.

Extending device **lifetime**
with improved software-based **heat** regulation.

What is DevOps++?

Governance Built-in

SOLID, code smells, (**energy**) efficiency checks
embedded in CI/CD.

What is DevOps++?

Systemic Optimization

Tie (**energy**) performance to business value & user impact

We have 10 core
principles to detect
these actions.

Idle Compute Reduction

Ghost energy, monitoring shuts it down.

CPU Load vs Business Value

Not every spike is user demand.

Right-Size Memory

Runtime feedback updates IaC.

IO Visibility

Stop blaming CPUs, fix DB locks.

Validate Early

Perf testing in PRs.

Prune Data

Shrink microservice / serverless payloads.

Compute Location

Edge, client, cloud — choose wisely.

Lightweight execution

WebAssembly.

Performance not by Perception

New nano/millisecond thresholds.

Efficiency in User Terms

OPUA metric (operations per user action).

It's all about
the playbook.

DevOps++ Playbook

Detect

 AI + observability → find **underutilization, bottlenecks, waiting**.

Expose

 Integration tests → make inefficiencies **visible (perf + energy)**.

Fix

 CI/CD auto-remediation (IaC + code).

Validate

 Ops + FinOps check in **acceptance & production**.

How can coding
practices reduce
energy consumption?

GreenCoding +

Overview

Boards

Repos

Pipelines

Pipelines

Environments

Library

Test Plans

Artifacts

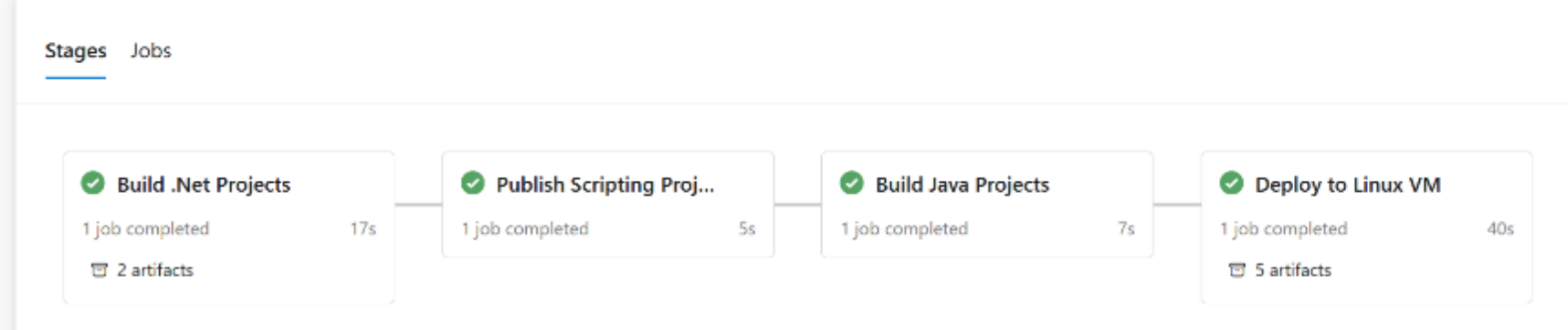
#20250130.41 • 1mln records
Green Coding

This run is being retained as one of 3 recent runs by pipeline.

Summary Code Coverage

Manually run by [REDACTED]

Repository and version	Time started and elapsed	Related	Tests and c
GreenCoding	30 Jan at 16:23	0 work items	Get star
matthijs 4e183ab7	1m 33s	7 published	



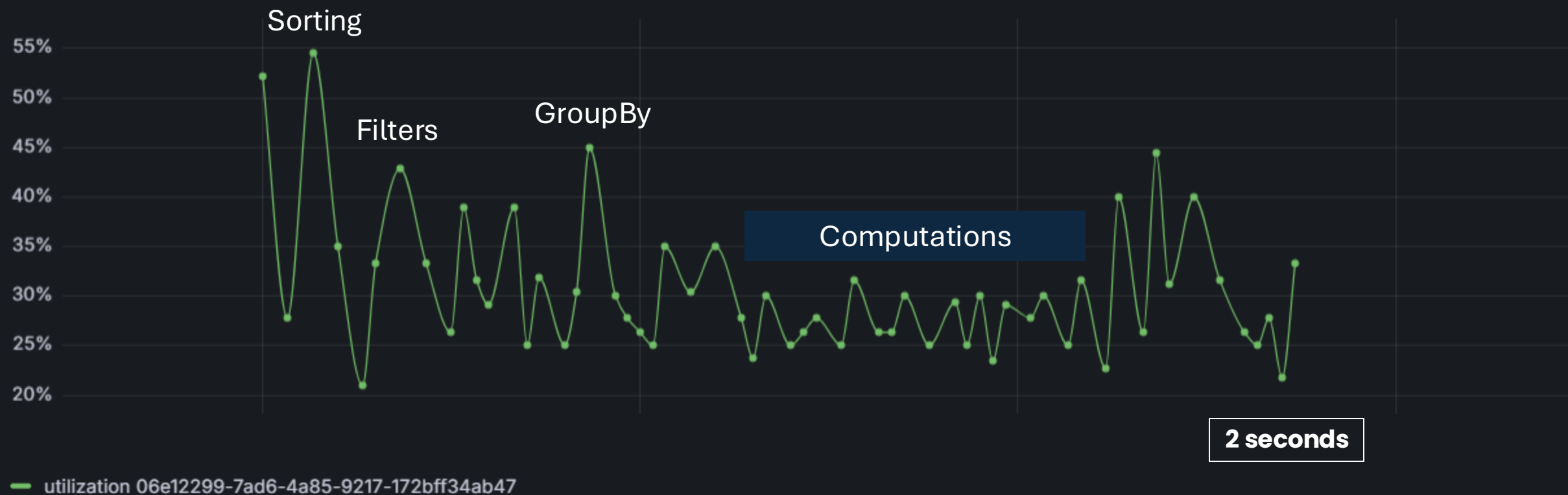
Published

Name	Size	
>  biodiversity-dotnet-artifact	6 MB	
>  biodiversity-java-artifact	56 MB	
>  biodiversity-nodejs-artifact	9 KB	
>  biodiversity-python-artifact	14 KB	
>  CodeTraceLogJava	651 B	
>  EnergyConsumptionReport	176 KB	
>  equal-power-tracer-app-artifact	2 MB	
>  ImpactFrameworkSciOutput	2 KB	
>  Prometheus output	161 KB	⋮
>  SCI Report	3 KB	

<https://localhost/api/GetInfo>

Utilization

Over time



Again, how can
coding practices
reduce energy
consumption?

As an example,
you are currently **working with**
the open-source **Express.js**
development team.

Total downloads of
Express.js is 5 to 10 Billion+

github.com/expressjs/express?utm_source=chatgpt.com

READMECode of conductContributingMIT licenseSecurity



Fast, unopinionated, minimalist web framework for [Node.js](#).

This project has a [Code of Conduct](#).

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 - [Security Issues](#)
 - [Running Tests](#)
- [Current project team members](#)
 - [TC \(Technical Committee\)](#)
 - [TC emeriti members](#)
 - [Triagers](#)
 - [Emeritus Triagers](#)

Packages

No packages published

Contributors337



[+ 323 contributors](#)

Languages

JavaScript99.9%Other0.1%

HighTech Innovators

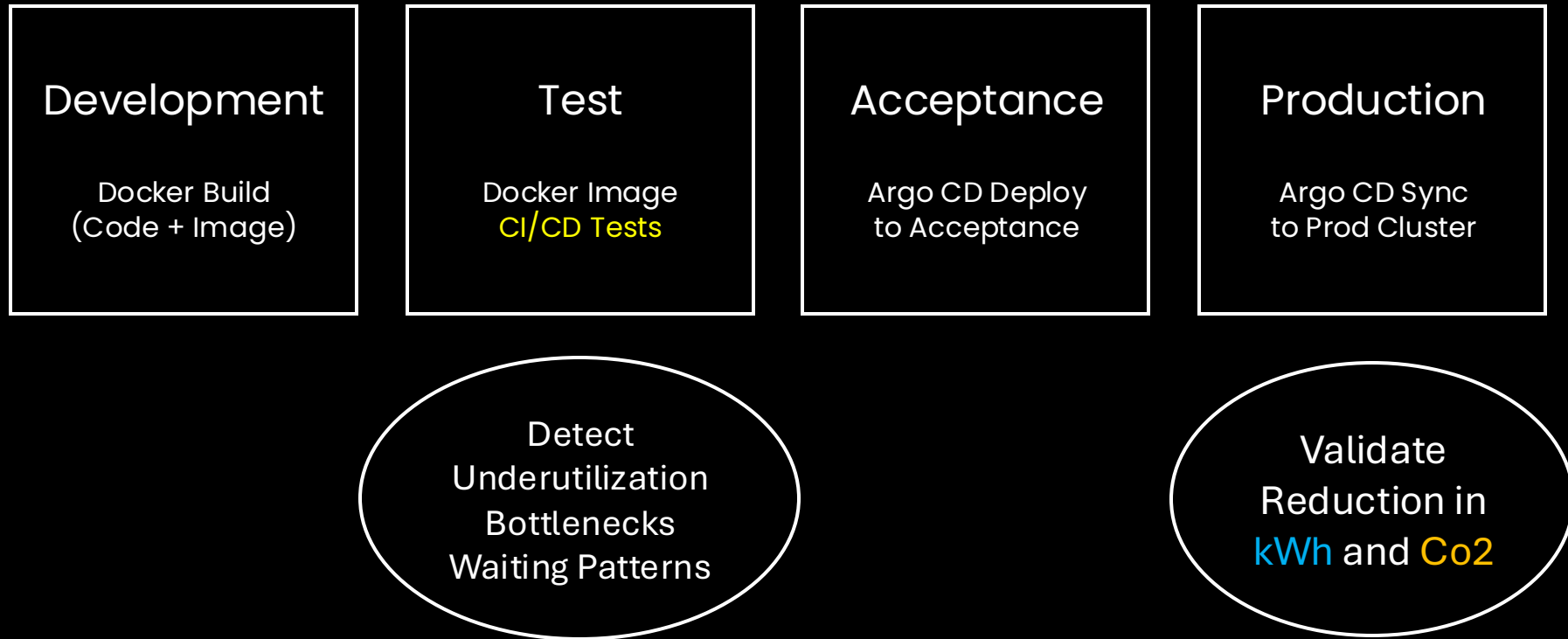
© 2024 HighTech Innovators. All rights reserved.

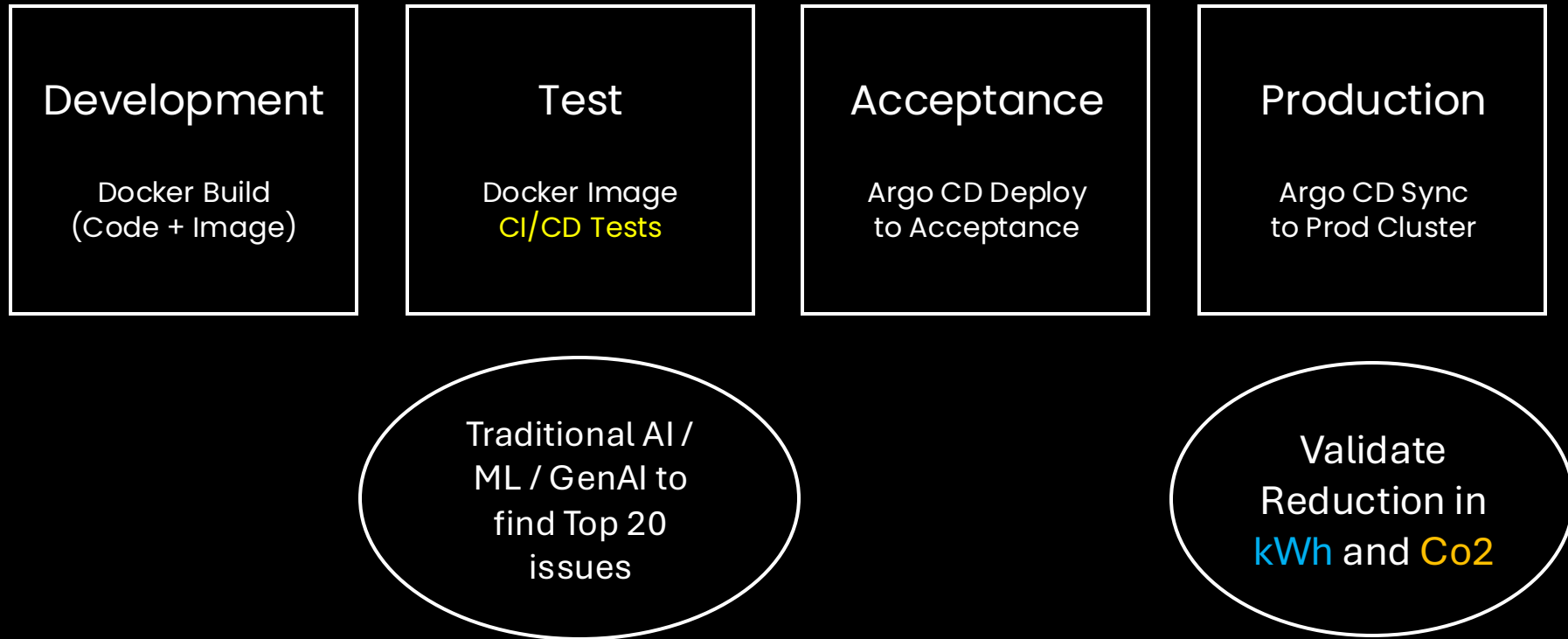
47

Nothing to improve
right?

Live Demo

How does testing
contribute to greener
software?





Fixing Friction: Ten Example Principles for DevOps++

Wilco Burggraaf · 4 min read · Jul 13, 2025



How Big Companies Can Cut Costs and Carbon Through Smarter Software Operations



In the modern enterprise, DevOps is no longer just about delivering software fast. It's about delivering software that runs efficiently, scales with purpose, and emits as little waste — digital, financial, and environmental — as possible. When you step inside large organizations with sprawling Kubernetes clusters, hybrid clouds, and thousands of containerized workloads, the patterns of waste are not always visible. But they are absolutely measurable. They sit in logs, dashboards, forgotten queues, and over-provisioned clusters.

This article lays out ten working principles — living examples that guide the evolving DevOps++ approach. These are not commandments or fixed rules. They are practical foundations, shaped by experience, and intended to be iterated upon by the community. DevOps++ itself is not a product, not a certification, and not a proprietary framework. It's an open, shared language for building software systems that are efficient, transparent, and resilient. It integrates development, infrastructure-as-code, and architecture into a unified rhythm where every change is an opportunity to lower cost, reduce footprint, and create more durable value.

Principle 1: Eliminate Idle Compute

Idle compute remains one of the most neglected drains in large environments. Containers spin in staging, dev clusters go forgotten, and workloads stay warm long past their use. DevOps++ bakes runtime activity checks into CI/CD workflows, ensuring policies are triggered to downscale or shut down compute when it no longer generates value. This is not about setting timers. It's about using application signals to guide compute lifecycles.

Principle 2: Challenge CPU Load Without Business Context

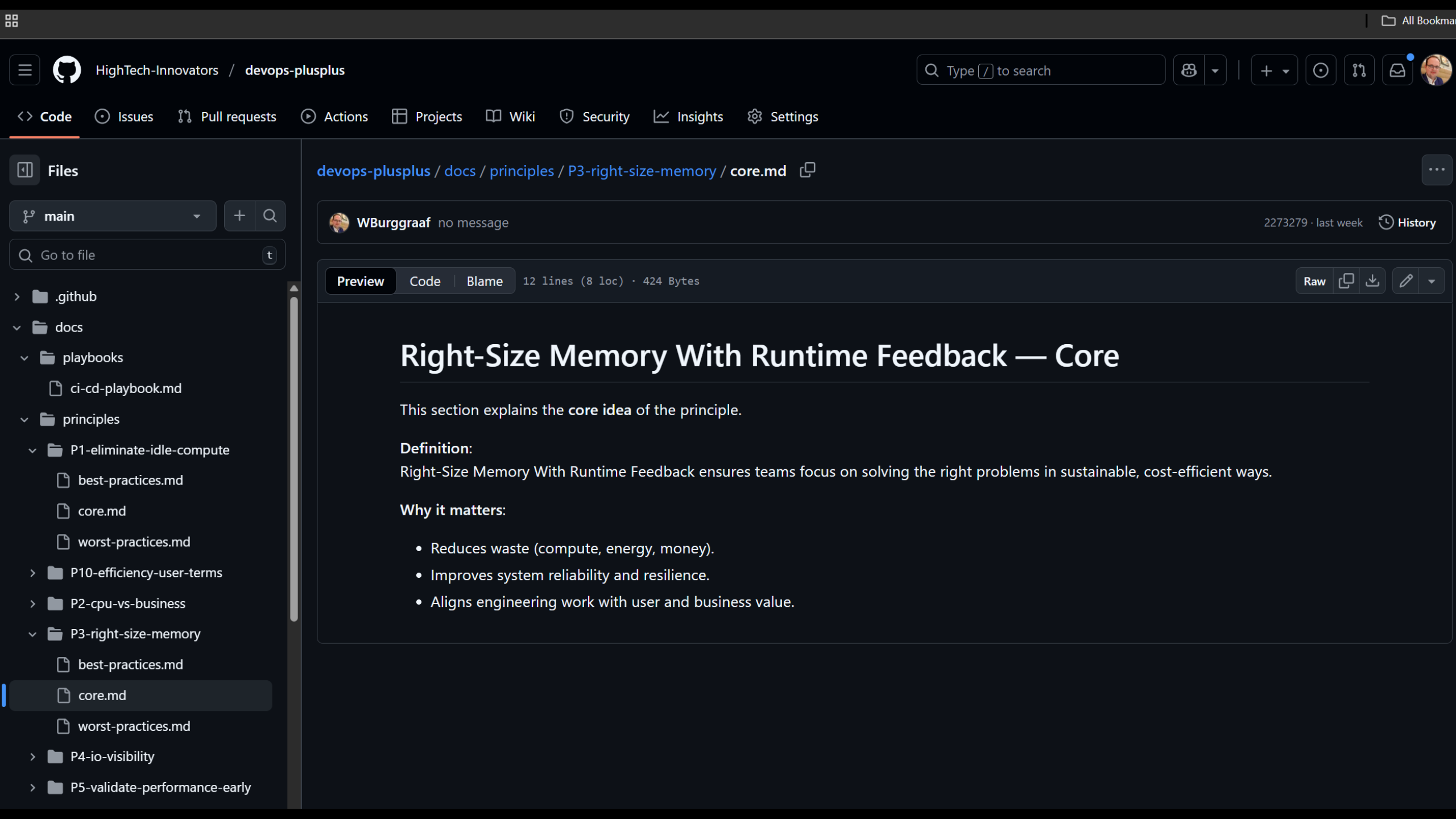
High CPU usage has long been mistaken as a symptom of scale. But not all scale is business value. In a DevOps++ world, teams trace CPU utilization back to business logic. Was that spike user-driven? Was it caused by inefficient polling? Every autoscaling policy is reviewed not by thresholds alone, but by relevance to product and impact on users.

Principle 3: Right-Size Memory With Runtime Feedback

No team gets memory sizing right the first time. That's why DevOps++ encourages memory observability to be integrated into CI test runs. Teams use historical runtime traces to automatically update IaC files with recommended limits. The result is fewer OOM crashes and far less memory waste.

Principle 4: Prioritize IO Visibility Over Compute Assumptions

Slow systems are often misunderstood. More cores are added, only to find the issue lives in disk IO, database locks, or blocked sockets. DevOps++ insists on full-stack latency tracing embedded into every deployment flow. Bottlenecks are located where they are — not where dashboards are easiest to read.



Files

main

Go to file

> .github

> docs

> playbooks

ci-cd-playbook.md

> principles

P1-eliminate-idle-compute

best-practices.md

core.md

worst-practices.md

> P10-efficiency-user-terms

> P2-cpu-vs-business

> P3-right-size-memory

best-practices.md

core.md

worst-practices.md

> P4-io-visibility

> P5-validate-performance-early

devops-plusplus / docs / principles / P3-right-size-memory / core.md

WBurggraaf

no message

2273279 · last week

History

Preview

Code

Blame

12 lines (8 loc) · 424 Bytes

Raw

Right-Size Memory With Runtime Feedback — Core

This section explains the **core idea** of the principle.

Definition:

Right-Size Memory With Runtime Feedback ensures teams focus on solving the right problems in sustainable, cost-efficient ways.

Why it matters:

- Reduces waste (compute, energy, money).
- Improves system reliability and resilience.
- Aligns engineering work with user and business value.